

**Table 1 Sample HVAC System Analysis and Selection Matrix (0 to 10 Score)**

| Goal: Furnish and install an HVAC system that provides moderate space temperature control with minimum humidity control at an operating budget of 220 kW/m <sup>2</sup> per year                                                                                                                                                                                                                                     |           |           |           |         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|---------|
| Categories                                                                                                                                                                                                                                                                                                                                                                                                           | System #1 | System #2 | System #3 | Remarks |
| 1. Criteria for selection: <ul style="list-style-type: none"> <li>25.6°C space temperature with <math>\pm 1.7</math> K control during occupied cycle, with 40% rh and <math>\pm 5\%</math> rh control during cooling.</li> <li>20°C space temperature with <math>\pm 1</math> K, with 20% rh and <math>\pm 5\%</math> rh control during heating season.</li> <li>First cost</li> <li>Equipment life cycle</li> </ul> |           |           |           |         |
| 2. Important factors: <ul style="list-style-type: none"> <li>First-class office space stature</li> <li>Individual tenant utility metering</li> </ul>                                                                                                                                                                                                                                                                 |           |           |           |         |
| 3. Other goals: <ul style="list-style-type: none"> <li>Engineered smoke control system</li> <li>ASHRAE <i>Standard</i> 62.1 ventilation rates</li> <li>Direct digital control building automation</li> </ul>                                                                                                                                                                                                         |           |           |           |         |
| 4. System constraints: <ul style="list-style-type: none"> <li>No equipment on first floor</li> <li>No equipment on ground adjacent to building</li> </ul>                                                                                                                                                                                                                                                            |           |           |           |         |
| 5. Energy use as predicted by use of an industry-acceptable computerized energy model                                                                                                                                                                                                                                                                                                                                |           |           |           |         |
| 6. Other constraints: <ul style="list-style-type: none"> <li>No perimeter finned-tube radiation or other type of in-room equipment</li> </ul>                                                                                                                                                                                                                                                                        |           |           |           |         |
| TOTAL SCORE                                                                                                                                                                                                                                                                                                                                                                                                          |           |           |           |         |

- System integration with other building systems
- Other HVAC goals
- Security and environmental health and safety criteria
- Building envelope
- Project timeline (design, equipment delivery, commissioning, training, and construction)
- Basis of design
- HVAC system analysis and selection matrix
- System narratives
- Budget costs (first cost, operating cost, and energy cost)
- Final recommendation(s)

A brief outline of each of the final selections should be provided along with overall budget and/or cost per square metre. In addition, HVAC systems deemed inappropriate should be noted as having been considered but not found applicable to meet the owner's primary HVAC goal.

The report should include an HVAC system selection matrix that identifies the one or two suggested HVAC system selections (primary and secondary, when applicable), system constraints, and other constraints and considerations. In completing this matrix assessment, the design engineer should identify the owner's input to the analysis. This input can also be applied as weighted multipliers, because not all criteria carry the same weighted value.

Many grading methods are available to complete an analytical matrix analysis. Probably the simplest is to rate each item excellent, very good, good, fair, or poor. A numerical rating system such as 0 to 10, with 10 equal to excellent and 0 equal to poor or not applicable, can provide a quantitative result. The HVAC system with the highest numerical value then becomes the recommended system to accomplish the goal.

The system selection report should include a summary followed by a more detailed account of the HVAC system analysis and system selection. This summary should highlight key points and findings that led to the recommendation(s). The analysis should refer to the system selection matrix (such as in Table 1) and the reasons for scoring.

With each HVAC system considered, the design engineer should note the criteria associated with each selection. Issues such as close-tolerance temperature and humidity control may eliminate some HVAC systems from consideration. System constraints, noted with each analysis, may eliminate potential HVAC systems. Advantages and disadvantages of each system should be noted with the scoring from the HVAC system selection matrix. This process should reduce HVAC selection to one or two optimum choices for presentation to the owner. Examples of similar installations for other owners can be included with this report to support the final recommendation. Identifying a third party for an endorsement allows the owner to inquire about the success of other HVAC installations.

## 2. HVAC SYSTEMS AND EQUIPMENT

Many built, expanded, and/or renovated buildings may be ideally suited for decentralized HVAC systems, with equipment located in, throughout, adjacent to, or on top of the building. The alternative to this decentralized approach is to use primary equipment located in a central plant (either inside or outside the building) with water and/or air required for HVAC needs distributed from this plant.

### Decentralized System Characteristics

The various types of decentralized systems are described in Chapter 2. The common element is that the required cooling is distributed throughout the building, generally with direct-expansion cooling, electric, or gas-fired heating of air systems, and on occasion, oil-fired warm-air heating units.

**Temperature, Humidity, and Space Pressure Requirements.** A decentralized system may be able to fulfill any or all of these design parameters, but typically not as efficiently or as accurately as a central system.

**Capacity Requirements.** A decentralized system usually requires each piece of equipment to be sized for zone peak capacity, unless the systems are variable-volume. Depending on equipment

construction or renovation because most codes prohibit combustible material in a ceiling return air plenum. Ceiling plenums and raised floors can also be used for supply air displacement systems to minimize horizontal distribution, along with other features discussed in Chapter 4.

Some ceiling plenum applications with lay-in panels do not work well where the stack effect of a high-rise building or high-rise elevators creates a negative pressure. If the plenum leaks to the low-pressure area, tiles may lift and drop out when the outside door is opened and closed.

The air temperature in a return air plenum directly below a roof deck is usually higher by 2 to 3 K during the air-conditioning season than in a ducted return. This can be an advantage to the occupied space below because heat gain to the space is reduced. Conversely, return air plenums directly below a roof deck have lower return air temperatures during the heating season than a ducted return and may require supplemental heat in the plenum.

Using a raised floor for air distribution is popular for computer rooms and cleanrooms, and is now being used in other HVAC applications. Underfloor air displacement (UFAD) systems in office buildings use the raised floor as a supply air plenum, which could reduce overall first cost of construction and ongoing improvement costs for occupants. Making the underfloor supply air plenum as airtight as possible is critical to the success of this type of plenum. This UFAD system improves air circulation to the occupied area of the space. See Chapter 20 of the 2023 *ASHRAE Handbook—HVAC Applications* and Chapter 20 of the 2021 *ASHRAE Handbook—Fundamentals* for more information on displacement ventilation and underfloor air distribution.

## 5. PIPE DISTRIBUTION

Piping should deliver refrigerant, heating water, chilled water, condenser water, fuel oil, gas, steam, and condensate drainage and return to and from HVAC equipment as directly, quietly, and economically as possible. Structural features of the building generally require mechanical and electrical coordination to accommodate P-traps, pipe pitch-draining of low points in the system, and venting of high points. When assessing application of pipe distribution to air distribution, the floor-to-floor height requirement can influence the pipe system: it requires less ceiling space to install pipe when compared to air ductwork. An alternative to horizontal piping is vertical pipe distribution, which may further reduce floor-to-floor height criteria. See Chapter 22 of the 2021 *ASHRAE Handbook—Fundamentals* for pipe sizing.

### Pipe Systems

HVAC piping systems can be divided into two parts: (1) piping in the central plant equipment room and (2) piping required to deliver refrigerant, heating water, chilled water, condenser water, fuel oil, gas, steam, and condensate drainage and return to and from decentralized HVAC and process equipment throughout the building. Chapters 11 to 15 discuss piping for various heating and cooling systems. Chapters 1 to 4 of the 2022 *ASHRAE Handbook—Refrigeration* discuss refrigerant system design and practices.

### Pipe Insulation

In new construction and renovation projects, most HVAC piping must be insulated. ASHRAE *Standard* 90.1 and Chapter 23 of the 2021 *ASHRAE Handbook—Fundamentals* have information on insulation and calculation methods. In most applications, water-source heat pump loop piping and condenser water piping may not need to be insulated.

## 6. SECURITY AND ENVIRONMENTAL HEALTH AND SAFETY

Since September 11, 2001, much attention has been given to protecting buildings against terrorist attacks via their HVAC systems. The first consideration should be to perform a risk assessment of the particular facility, which may be based on usage, size, population, and/or significance. The risk assessment is a subjective judgment by the building owner (and sometimes by a registered/certified security professional) of whether the building is at low, medium, or high risk. An example of low-to-medium risk buildings may be suburban office buildings, shopping malls, hospitals, educational institutions, or major office buildings. High-risk buildings may include major government buildings. The level of protection designed into these buildings may include enhanced particulate filtration, gaseous-phase filtration, and various control schemes to allow purging of the facility using either the HVAC system or an independent dedicated system.

Enhanced particulate filtration for air-handling systems to the level of MERV 14 to 16 filters not only tends to reduce circulation of dangerous substances (e.g., anthrax), but also provides better indoor air quality (IAQ). Gaseous-phase filtration can remove harmful substances such as sarin and other gaseous threats. For buildings of all risk levels, consideration should include proper location of outdoor air intakes to account for other security concerns, such as toxic fumes from a local accident (e.g., punctured tank from a nearby railroad car or tanker truck). The extent to which the HVAC system designer should use these measures could depend on the recommendations of an experienced security and environmental health and safety consultant or agency.

In any building, consideration should be given to protecting outdoor air intakes against insertion of dangerous materials by locating the intakes on the roof or substantially above grade level. Separate systems for mailrooms, loading docks, and other similar spaces should be considered so that any dangerous material received cannot be spread throughout the building from these types of vulnerable spaces. Emergency ventilation systems for these types of spaces should be designed so that upon detection of suspicious material, toxic fumes, or chemical spill, these spaces can be quickly purged and maintained at a negative pressure.

A more extensive discussion of this topic can be found in ASHRAE's *Guideline* 29. Note, however, that fan energy required to move air through these high efficiency filters is very significant.

## 7. AUTOMATIC CONTROLS AND BUILDING MANAGEMENT SYSTEMS

Basic HVAC system controls are primarily direct digital controls (DDC), but there are also electric, electronic, and pneumatic controls (found largely as part of existing building HVAC systems). Depending on the application, the design engineer may recommend a simple and basic system strategy as a cost-effective solution to an owner's heating, ventilation, and cooling needs. Chapter 48 of the 2023 *ASHRAE Handbook—HVAC Applications* and Chapter 7 of the 2021 *ASHRAE Handbook—Fundamentals* discuss automatic control in more detail.

The next level of HVAC system management is DDC wireless systems. Many DDC installations will use either pneumatic or electric control damper and valve actuators. This automatic control enhancement may include energy monitoring and energy management software. Controls may also be accessible by the building manager using a modem to a remote computer at an off-site location or via a smartphone or tablet computer with Internet connection to the control system. Building size has little to no effect on modern computerized controls: programmable controls can be furnished on the smallest HVAC equipment for the smallest projects. Chapter 43

ures 25 and 26). Typically, authorities in the range of 30 to 50% (for piping systems with 2:1 branch-to-riser pressure drop ratios) are adequate for HVAC coil applications, with higher  $\Delta t$  coils requiring less valve authority. Authorities less than 30% should be avoided because they lead to a flow characteristic shift that makes the equal-percentage valve appear to be more linear in nature (Figures 27 and 28). For the three-way valve shown in Figure 23, the full-open pressure drop should be from C to D. The pressure drop in the bypass balancing valve in the three-way valve circuit should be set to equal that in the coil (load), Chapter 47 discusses valve authority in three way modulating applications.

Control valves should be sized on the basis of the required valve coefficient  $C_v$  for the required pressure drop and flow. For more information, see the section on Control Valve Sizing under Automatic Valves in Chapter 47. Briefly, in variable-speed pumping systems using differential pressure control of the branch, the pressure drop of concern is the controlled differential, and is generally placed to measure the valve and coil pressure drops. The valve pressure drop for control, then, is that required to provide complementary characteristic. In the previous coils, a 6.7 K  $\Delta t$  was used, and valve authority of 50 to 100% provides good results, implying a pressure drop on the valve equal to that of the coil. Control valves that follow are thus dependent on the controlled hydraulic performance. In these valves, attention must also be paid to valve construction (especially the maximum pressure allowed on the valve body, and maximum allowed pressure drop across the valve).

If a system is to be designed with multiple zones of control such that load response is to be by constant flow through the load and variable  $\Delta t$ , control cannot be achieved by valve control alone; a load pump is required.

Several control arrangements of load pump and control valve configurations are shown in Figure 29. Note that, in all three configurations, the common pipe has no restriction or check valve. In all configurations, there is no difference in control as seen by the load. However, the basic differences in control are

- With the two-way valve configuration (Figure 29A), the distribution system sees variable flow and constant  $\Delta t$ , whereas with both three-way configurations, the distribution system sees constant flow and variable  $\Delta t$ .
- Configuration B differs from C in that the pressure required through the three-way valve in Figure 29B is provided by the load pump, whereas in Figure 29C it is provided by the distribution pump(s).

### Alternatives to Control Valves

Use of variable-frequency drives has greatly increased because advances in technology and decreases in cost have made these drives an attractive alternative to using control valves for heat transfer control (Figure 30).

Green (1994) tested the control stability of variable-speed circulating pumps compared to control valves. Stability of the controlled

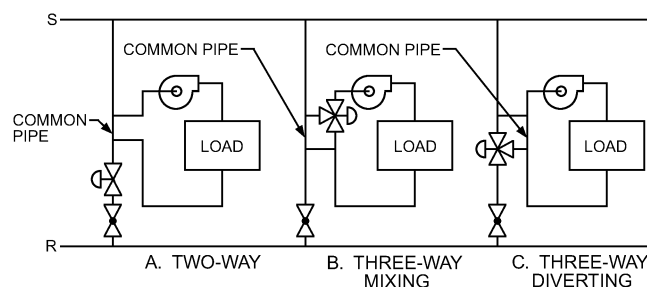


Fig. 29 Load Pumps with Valve Control

discharge temperature of the coil was on the order of  $\pm 0.28$  K. Large coils with pump and separate drive may also be economical compared to similarly sized valves and actuators. The stability of on/off pump control was also found to be reasonable. The attraction of a pump over a control valve is the reduction of pressure and thus energy used by the distribution pumping system. For the valve to work properly, pump pressure must be throttled to control flow. Any throttling process is inherently energy inefficient. Direct load control by a pump provides only the energy required to overcome the friction loss of the load heat transfer device and piping. Properly sequenced, the distribution pumping system can reduce flow, which causes higher coil entering water temperatures, which causes the circulating pump to operate at a higher flow. This is helpful in allowing for higher percentages of heat transfer capability of the sensible and latent loads through the coil. Applications of this control method must also take into account the potential of gravity flow in the piping scheme.

### 2.6 LOW-TEMPERATURE HEATING SYSTEMS

These systems are used for heating spaces or processes directly, as with standing radiation and process heat exchangers, or indirectly, through air-handling unit coils for preheating, for reheating, or in hot-water unit heaters. They are generally designed with supply water temperatures from 80 to 115°C and temperature drops from 10 to 55 K.

In the United States, hot-water heating systems were historically designed for a 93°C supply water temperature and an 11 K temperature drop. This practice evolved from earlier gravity system designs and provides convenient design relationships for heat transfer coefficients related to coil tubing and finned-tube radiation and for calculations. Because many terminal devices still require these flow rates, it is important to recognize this relationship in selecting devices and designing systems.

However, the greater the temperature range (and related lower flow rate) that can be applied, the less costly the system is to install and operate. A lower flow rate requires smaller and less expensive piping, less secondary building space, and smaller pumps. Also, smaller pumps require less energy, so operating costs are lower.

### Nonresidential Heating Systems

Possible approaches to enhancing the economics of large heating systems include (1) higher supply temperatures, (2) primary/secondary pumping, and (3) terminal equipment designed for smaller flow rates. The three techniques may be used either singly or in combination.

Using higher supply water temperatures achieves higher temperature drops and smaller flow rates. Terminal units with a reduced heat-

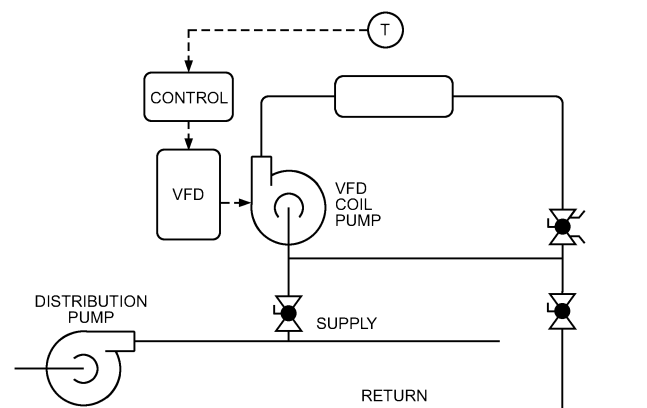


Fig. 30 Schematic of Variable-Speed Pump Coil Control

air distribution systems (i.e., those with supply air temperature below 11°C). These outlets include special features and rapidly mix cold supply air with room air at the outlet and effectively reduce the temperature differential between the supply and room air. For further information, designers can consult ASHRAE's (1996) *Cold Air Distribution System Design Guide*.

Accessories used with an outlet can regulate the volumetric flowrate of supply air and control its flow pattern. Accessories may also be necessary for proper air distribution in a space, so they must be selected and used according to the manufacturers' recommendations.

### Factors That Influence Selection

**Coanda (Surface or Ceiling) Effect.** Via restricting secondary air entrainment, a jet moving adjacent to or in contact with an impermeable wall or ceiling creates a low-pressure area immediately adjacent to that surface, causing the air to remain in contact with the surface substantially throughout the length of throw. This **Coanda effect**, also referred to as the **surface or ceiling effect**, counteracts, to varying extents, the negative-buoyancy-driven drop of a horizontally projected cool airstream.

Round and four-way horizontal-throw ceiling outlets exhibit a high Coanda effect because the discharge air pattern blankets the ceiling area surrounding each outlet. This effect diminishes with a directional discharge that does not blanket the full ceiling surface surrounding the outlet due to entrainment into the exposed sides of the jet. Sidewall outlets exhibit varying degrees of Coanda effect, depending on the spread of the particular air pattern, the angle of the jet, and the distance from the outlet's discharge to the surface.

When outlets are mounted such that they discharge into a free space not near an adjacent surface (e.g., exposed duct-mounted supply air grille applications), the airstream entrains air around the entire perimeter of the jet. As a result, the rate of entrainment is higher and the throw is shortened. More information on jet behavior can be found in Chapter 20 of the 2021 *ASHRAE Handbook—Fundamentals*.

**Temperature Differential.** The greater the temperature differential between the supply air projected into a space and the air in the space, the greater the buoyancy effect on the path of the supply airstream. Heated, horizontally projected air rises and cooled air falls, so this effect should be considered during outlet selection. Chapter 20 of the 2021 *ASHRAE Handbook—Fundamentals* has further discussion of the buoyancy effect.

Low-temperature supply air or cold building start-up in a humid environment may cause condensation. Consider the effect of condensation on outlet and space surfaces during outlet selection.

**Sound Level.** The sound level from an outlet is largely a function of its discharge velocity and transmission of HVAC system noise, such as from fans and air turbulence, and may be desirable for masking other sounds, or, if too high, may be considered noise by occupants. For a given air capacity, a larger outlet face area has a lower discharge velocity and correspondingly lower generated sound. However, a larger outlet also allows more system-generated sound to pass through the outlet, which may appear as outlet-generated noise to occupants. High-frequency noise can result from excessive outlet velocity but may also be generated in the duct by the moving airstream. Low-frequency noise is usually generated by mechanical equipment and transmitted through the duct and outlet to the room.

The cause of a noise complaint can usually be pinpointed as outlet or system sounds by removing the outlet's core during operation. If the noise remains essentially unchanged, the system is the source. If the noise is significantly reduced, the outlet is likely the source. The noise may be caused by a highly irregular velocity profile at the entrance to the outlet. ASHRAE research project RP-1335 (Landsberger 2011) and Chapter 58 of the 2023 *ASHRAE Handbook—*

*HVAC Applications* have more details on the effects of inlet conditions on air outlets.

**Variable-Air-Volume (VAV) Outlet.** A VAV outlet is an air outlet that self-modulates its primary air flow rate with an integral air damper. Select outlets based on the total range of airflow needed for the space served. Outlet performance characteristics should be evaluated at both minimum and maximum flow. More information regarding selection of outlets can be found in Chapter 58 of the 2023 *ASHRAE Handbook—HVAC Applications*.

### Outlet Selection Procedure

The following procedure is generally used in selecting and locating an outlet in a fully mixed system. More details and examples are available in Rock and Zhu (2002) and Chapter 58 of the 2019 *ASHRAE Handbook—HVAC Applications*.

1. Determine the design flow rates of air to be supplied to each room. (See Chapters 17 and 18 of the 2021 *ASHRAE Handbook—Fundamentals* to determine air quantities for heating and cooling modes.)
2. Select the type and quantity of outlets for each room, considering factors such as air quantity required, distance available for throw or radius of diffusion, structural characteristics, and architectural concepts.
3. Outlets may be sized and located to distribute air in the space to achieve acceptable temperature and velocity in the occupied zone. Select and locate outlets such that the air supplied reaches the breathing zone, as defined by ASHRAE *Standard* 62.1 for heating and cooling in many buildings.
4. Select the proper size outlet from the manufacturers' performance data according to air quantity, neck and discharge velocity, throw, distribution pattern, and sound level. Note manufacturers' recommendations regarding use. Also, obstructions to the primary air distribution pattern require special study. In an open space, the interaction of airstreams from multiple air outlets may alter a single outlet's throw, air temperature, or air velocity. As a result, manufacturers' data may be insufficient to predict air motion in a particular space. The **air diffusion performance index (ADPI)** is a useful tool in predicting outlet performance for fully mixed systems; see Chapter 58 of the 2023 *ASHRAE Handbook—HVAC Applications* for more information.

## 2.2 FULLY STRATIFIED SYSTEMS

Stratified room air distribution systems, typically called **thermal displacement ventilation**, generally rely on supply outlets with very low discharge velocities (e.g., 0.2 to 0.36 m/s based on total face area) to produce minimal room air entrainment so that much of the temperature difference between supply and the occupied zone's air is preserved. With outlets placed low, and when in cooling mode, cool supply air accumulates near the floor. Heat sources in the occupied zone create convection plumes that originate around the boundaries of the heat source and rise naturally because of their buoyancy. If these sources are near the supply airstream, supply air is entrained to fill the void of the rising convection plume.

The cooling-mode temperature differential between supply and room air is generally less than that commonly used for fully mixed systems. Exhaust air is generally higher temperature compared to fully mixed systems, especially with high ceilings. Stratified systems used in transient spaces where thermal comfort is less critical may, however, use air temperature differentials like those in fully mixed systems.

### Factors that Influence Selection

**Space Considerations.** To maximize system efficiency, supply outlets are typically located below the occupant breathing level, so

Coil ratings and selections can be obtained from manufacturers' catalogs. Most catalogs contain extensive tables giving the performance of coils at various air and water velocities and entering humidity and temperatures. Most manufacturers provide computerized coil selection programs to potential customers. The final choice can then be made based on system performance and economic requirements.

### Performance and Ratings

The long-term performance of an extended-surface air-cooling and dehumidifying coil depends on its correct design to specified conditions and material specifications, proper matching to other system components, proper installation, and proper maintenance as required.

In accordance with AHRI *Standard* 410, dry-surface (sensible cooling) coils and dehumidifying coils (which both cool and dehumidify), particularly those for field-assembled coil banks or factory-assembled packaged units using different combinations of coils, are usually rated within the following parameters:

Entering air dry-bulb temperature: 18 to 38°C

Entering air wet-bulb temperature: 15 to 30°C (if air is not dehumidified in the application, select coils based on sensible heat transfer)

Air face velocity: 1 to 4 m/s

Evaporator refrigerant saturation temperature: –1 to 13°C at coil suction outlet (refrigerant vapor superheat at coil suction outlet is 3 K or higher)

Entering chilled-water temperature: 1.5 to 18°C

Water velocity: 0.3 to 2.4 m/s

For ethylene glycol solution: 0.3 to 1.8 m/s, –18 to 32°C entering dry-bulb temperature, 15 to 27°C entering wet-bulb temperature, 10 to 60% aqueous glycol concentration by mass

The air-side ratio of sensible to total heat removed by dehumidifying coils varies in practice from about 0.6 to 1.0 (i.e., sensible heat is from 60 to 100% of the total, depending on the application). For information on calculating a dehumidifying coil's sensible heat ratio, see the section on Performance of Dehumidifying Coils, or Appendix D of *Cooling and Heating Load Calculation Principles* (Pedersen et al. 1998). For a given coil surface design and arrangement, the required sensible heat ratio may be satisfied by wide variations in and combinations of air face velocity, in-tube temperature, flow rate, entering air temperature, coil depth, etc., although the variations may be self limiting. The maximum coil air face velocity should be limited to a value that prevents water carryover into the air ductwork. Dehumidifying coils for comfort applications are frequently selected in the range of 2.0 to 2.5 m/s air face velocity.

Operating ratings of dehumidifying coils for factory-assembled, self-contained air conditioners are generally determined in conjunction with laboratory testing for the system capacity of the complete unit assembly. For example, a standard rating point has been 54 L/s per kilowatt of refrigeration effect, not to exceed 60 L/s per kilowatt for unitary equipment. Refrigerant (e.g., R-22) duty would be 3 to 6 K superheat for an appropriate balance at 7°C saturated suction. For water coils, circuitry would operate at 1.2 m/s, 5.5°C inlet water, 6.7 K rise (or 36 mL/s per kilowatt of refrigeration effect). The standard ratings at 26.7°C db and 19.5°C wb are representative of the entering air conditions encountered in many comfort operations. Although indoor conditions are usually lower than 19.5°C wb, it is usually assumed that introducing outdoor air brings the air mixture to the cooling coil up to about 26.7°C db/19.5°C wb entering air design conditions.

Dehumidifying coils for field-assembled projects and central station air-handling units were formerly selected according to coil rating tables but are now selected by computerized selection programs. Either way, selecting coils from the load division indicated by the load calculation works satisfactorily for the usual human comfort

applications. Additional design precautions and refinements are necessary for more exacting industrial applications and for all types of air conditioning in humid areas. One such refinement, the dual-path air process, uses a separate cooling coil to cool and dehumidify ventilation air before mixing it with recirculated air. This process dehumidifies what is usually the main source of moisture: makeup outdoor air. Condenser heat reclaim (when available) is another refinement required for some industrial applications and is finding greater use in commercial and comfort applications.

Airflow ratings are based on standard air of 1.20 kg/m<sup>3</sup> at 20°C and a barometric pressure of 101.325 kPa. In some mountainous areas with a sufficiently large market, coil ratings and altitude-corrected psychometrics are available for their particular altitudes.

When checking the operation of dehumidifying coils, climatic conditions must be considered. Most problems are encountered at light-load conditions, when the cooling requirement is considerably less than at design conditions. In hot, dry climates, where the outdoor dew point is consistently low, dehumidifying is not generally a problem, and the light-load design point condition does not pose any special problems. In hot, humid climates, the light-load condition has a higher proportion of moisture and a correspondingly lower proportion of sensible heat. The result is higher dew points in the conditioned spaces during light-load conditions unless a special means for controlling inside dew points (e.g., reheat or dual path) is used.

Fin surface freezing at light loads should be avoided. Freezing occurs when a dehumidification coil's surface temperature falls below 0°C. Freezing does not occur with standard coils for comfort installations unless the refrigerant evaporating temperature at the coil outlet is below –4 to –2°C saturated; the exact value depends on the design of the coil, its operating dew point, and the amount of loading. With coil and condensing units to balance at low temperatures at peak loads (not a customary design choice), freezing may occur when load suddenly decreases. The possibility of this type of surface freezing is greater if a bypass is used because it causes less air to be passed through the coil at light loads.

## 4. AIRFLOW RESISTANCE

A cooling coil's airflow resistance (air friction) depends on the tube pattern and fin geometry (tube size and spacing, fin configuration, and number of in-line or staggered rows), coil face velocity, and amount of moisture on the coil. The coil air friction may also be affected by the degree of aerodynamic cleanliness of the coil core; burrs on fin edges may increase coil friction and increase the tendency to pocket dirt or lint on the faces. A completely dry coil, removing only sensible heat, offers approximately one-third less resistance to airflow than a dehumidifying coil removing both sensible and latent heat.

For a given surface and airflow, increasing the number of rows or fins increases airflow resistance. Therefore, final selection involves economic balancing of the initial cost of the coil against the operating costs of the coil geometry combinations available to adequately meet the performance requirements.

The aluminum fin surfaces of new dehumidifying coils tend to inhibit condensate sheeting action until they have aged for a year. Hydrophilic aluminum fin surface coatings reduce water droplet surface tension, producing a more evenly dispersed wetted surface action at initial start-up. Manufacturers have tried different methods of applying such coatings, including dipping the coil into a tank, coating the fin stock material, or subjecting the material to a chemical etching process. Tests have shown as much as a 30% reduction in air pressure drop across a hydrophilic coil as opposed to a new untreated coil.

limited with vapor-charged thermostatic expansion valves. Low-discharge-pressure (low ambient) operation decreases pressure drop across valves and capillaries so that full flow is not maintained. Decreased capacity, low coil temperatures, and freeze-up can result unless low ambient condensing-pressure control is provided.

Properly designed unitary equipment allows only a minimum amount of liquid refrigerant to return through the suction line to the compressor during non-steady-state operation. Normally, the heat absorbed in the evaporator vaporizes all of the refrigerant and adds a few degrees of superheat. However, any conditions that increase refrigerant flow beyond the evaporator's heat transfer capabilities can cause liquid carryover into the compressor return line. This increase may be caused by a poorly positioned thermal element of an expansion valve or by an increase in the condensing pressure of a capillary system, which may be caused by fouled condenser surfaces, excessive refrigerant charge, reduced flow of condenser air or water, or the higher temperature of the condenser cooling medium. Heat transfer at the evaporator may be reduced by dirty surfaces, low-temperature air entering the evaporator, or reduced airflow caused by a blockage in the air system.

**Piping.** Transient flow conditions are a special concern. During *off* periods, refrigerant migrates and condenses in the coldest part of the system. In an air conditioner in a cooled space, this area is typically the evaporator. When the compressor starts, liquid tends to return to the compressor in slugs. The severity of **slugging** is affected by temperature differences, *off* time, component positions, and traps formed in suction lines. Various methods such as suction-line accumulators, specially designed compressors, the refrigerant pumpdown cycle, nonbleed port thermostatic expansion valves, liquid-line solenoid valves, or limited refrigerant charge are used to avoid equipment problems associated with excessive liquid return. Chapter 1 of the 2022 *ASHRAE Handbook—Refrigeration* has further information on refrigerant piping.

Strainers and filter-driers minimize the risk of foreign material restricting capillary tubes and expansion valves (e.g., small quantities of solder, flux, copper oxides, varnish). Overheated and oxidized oil may dissolve in warm refrigerant and deposit at lower temperatures in capillary tubes, expansion valves, and evaporators. Filter-driers are required by virtually all manufacturers when using hygroscopic lubricants such as polyol ester (POE) and polyvinyl ether (PVE), particularly for split-system units, to remove any moisture introduced during installation or servicing. Moisture contamination can cause oil breakdown, motor insulation failure, and freezing or other restrictions at the expansion device.

**Capacity Control.** Buildings with high internal heat loads require cooling even at low outdoor temperatures. The capacity of air-cooled condensers can be controlled by changing airflow or flooding tubes with refrigerant or using variable-speed compressors. Airflow can be changed by using dampers, adjusting fan speed, or stopping some of the fan motors in a multifan system.

In cool weather, air conditioners operate for short periods only. If the weather is also damp, high humidity levels with wide variances may occur. Properly designed capacity-controlled units operate for longer periods, which may improve humidity control and comfort. In any case, cooling equipment should not be grossly oversized.

Units with two or more separate refrigerant circuits allow independent operation of the individual systems, which reduces capacity while better matching changing load conditions. Larger, single-compressor systems may offer capacity reduction through unloading compressors, variable-speed or multispeed compressors, multiple compressors, or hot-gas bypass controls. At full-load operation, efficiency is unimpaired. However, reduced-capacity operation may increase or decrease system efficiency, depending on the capacity-reduction method used.

Variable-speed, multispeed, and staged multiple compressors can improve efficiency at part-load operation. Unloading can increase or

decrease system performance, depending on the particular method used. Multispeed and variable-speed compressors, multiple compressors, and unloading generally produce higher comfort levels through lower cycling and better matching of capacity to load. Hot-gas bypass does not reduce capacity efficiently, although it generally provides a wider range of capacity reduction. Units with capacity-reduction compressors usually have capacity-controlled evaporators; otherwise evaporator coil temperatures may be too high to provide dehumidification. Capacity-controlled evaporators are usually split, with at least one of the expansion valves controlled by a solenoid valve. Evaporator capacity is reduced by closing the solenoid valve. Compressor capacity-reduction controls or a hot-gas bypass system then provides maximum dehumidification, while the evaporator coil temperature is maintained above freezing to avoid coil frosting. Chapter 2 of the 2022 *ASHRAE Handbook—Refrigeration* contains details on hot-gas bypass.

## Air-Handling Systems

High airflow, low-static-pressure performance, simplicity, economics, and compact arrangement are characteristics that make propeller fans particularly suitable for nonducted air-cooled condensers. Small-diameter fans are direct-driven by four-, six-, or eight-pole motors. Low starting-torque requirements allow use of single-phase shaded pole and permanent split-capacitor (PSC) fan motors and simplify speed control for low-outdoor-temperature operation. Many larger units use multiple fans and three-phase motors. Larger-diameter fans are belt-driven at a lower speed to maintain low tip speeds and quiet operation.

Centrifugal blowers meet the higher static-pressure requirements of ducted air-cooled condensers, forced-air furnaces, and evaporators. Indoor airflow must be adjusted to suit duct systems and plenums while providing the required airflow to the coil. Some small blowers are direct driven with multispeed motors. In ductless indoor units, such as wall-mounted, under-ceiling-mounted, and ceiling cassette units, aluminum or fire-retardant plastic centrifugal fans are used. Large blowers are always belt driven and may have variable-pitch motor pulleys or a variable-frequency drive (VFD) for airflow adjustment. Vibration isolation reduces the amount of noise transmitted by bearings, motors, and blowers into cabinets. (See Chapter 21 for details of fan design and Chapters 19 and 20 for information on air distribution systems.)

PSC motors are widely applied to furnaces, air handlers, condensing units, and packaged equipment. They offer relatively low cost and high reliability, but efficiencies are relatively low. For applications with more stringent efficiency requirements, use electronically commutated motors (ECM). In addition to efficiency advantages, these motors can maintain a set airflow as static pressure varies. More recently, constant-torque motors (CTMs) have been used in indoor applications. These motors offer similar efficiencies as ECMs but at a lower cost. However, they maintain constant torque rather than constant airflow as static pressures change. This means that airflows vary as static pressure changes, but not as much as with a PSC motor. The fan motor selection process should consider the advantages and disadvantages of each motor technology, as well as the cost implications. Further information on motors used in HVAC applications may be found in Chapter 45.

Disposable fiberglass filters are popular because they are available in standard sizes at low cost. Cleanable filters can offer economic advantages, especially when cabinet dimensions are not compatible with common sizes. Charcoal filters can help remove volatile organic compounds (VOCs) or microparticles. Electronic or other high-efficiency air cleaners are used when a high degree of cleaning is desired. Larger equipment frequently is provided with automatic roll filters or high-efficiency bag filters. (See Chapter 29 for additional details about filters.)